

Practical 1

Title : Exploring Network Diagnostic Commands: ipconfig, ping, tracert, and nslookup

Aim: To understand basic network configuration and analyze network connectivity, routing, and DNS resolution using Windows Command Prompt utilities.

Objective: Understand how data flows over networks and how devices communicate.

Required Software

- Windows Command Prompt / PowerShell
- macOS/Linux Terminal
- Internet Connection

Theory:

Whenever we access a website such as www.google.com, several networking processes occur:

1. The computer checks its network configuration.
2. DNS converts the website name into an IP address.
3. Data packets travel through multiple routers.
4. The destination server replies to the request.

Network administrators use diagnostic commands to troubleshoot connectivity issues. In this experiment, students will explore four essential commands:

- ipconfig
- ping
- tracert
- nslookup

[Students need to write the explanation of each command below the screenshot of that particular command explaining all the terms like TTL, Time, etc.]

Part A – Exploring Network Configuration

Command

ipconfig

Purpose

Displays the current TCP/IP configuration of the computer.

Sample Output

```

PS C:\Users\shaya> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 16:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

    Connection-specific DNS Suffix  . : bbrouter
    Link-local IPv6 Address . . . . . : fe80::f787:ed7e:f581:793c%34
    IPv4 Address. . . . . : 192.168.1.8
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::9462:65ef:72df:3a52%62
    IPv4 Address. . . . . : 172.17.112.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

```

It shows the IPv4 Address, which identifies the device on the network, the Subnet Mask, which separates the network and host portions of the IP address, and the Default Gateway, which is the router used to access other networks, including the internet.

IPv4 Address : 192.168.1.25

Subnet Mask : 255.255.255.0

Default Gateway . . : 192.168.1.1

Part B – Testing Connectivity

Command

ping google.com

Purpose

Checks whether the destination device is reachable.

Sample Output

```
PS C:\Users\shaya> ping google.com

Pinging google.com [142.250.195.14] with 32 bytes of data:
Reply from 142.250.195.14: bytes=32 time=7ms TTL=118
Reply from 142.250.195.14: bytes=32 time=5ms TTL=118
Reply from 142.250.195.14: bytes=32 time=5ms TTL=118
Reply from 142.250.195.14: bytes=32 time=6ms TTL=118

Ping statistics for 142.250.195.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 7ms, Average = 5ms
```

Shows if computer can communicate with the Google server. Bytes indicates the size of the data packet sent (usually 32 bytes), Time shows the round-trip time taken for the packet to travel to the server and back (measured in milliseconds), and TTL (Time To Live) indicates the maximum number of network hops a packet can pass through before being discarded, helping prevent packets from circulating indefinitely.

Reply from 142.250.xx.xx

bytes=32

time=18ms

TTL=117

Part C – Tracing the Packet Route

Command

```
tracert google.com
```

Purpose

Displays the routers (hops) through which packets travel before reaching the destination.

Sample Output

```

PS C:\Users\shaya> tracert google.com

Tracing route to google.com [142.250.195.14]
over a maximum of 30 hops:

  1      2 ms      2 ms      1 ms  REALTEK [192.168.1.1]
  2     11 ms     12 ms     12 ms  172.25.1.118
  3     13 ms     11 ms     15 ms  172.25.1.117
  4      *        *        *      Request timed out.
  5      6 ms     11 ms     12 ms  26.188.100.175.mipl.com [175.100.188.26]
  6      8 ms      6 ms      5 ms  74.125.37.7
  7      9 ms      3 ms      5 ms  209.85.248.61
  8      6 ms      4 ms      7 ms  del12s09-in-f14.1e100.net [142.250.195.14]

Trace complete.

```

This command is used to trace the path that data packets take from my computer to the Google server. It displays each router the packets pass through along the way and shows the time taken to reach each hop(the destination needed to reach).

- 1 1 ms Router
- 2 5 ms ISP Router
- 3 12 ms Regional Router
- 4 20 ms Google Network

Part D – Domain Name Resolution

Command

```
nslookup google.com
```

Purpose

Queries the DNS server to convert a domain name into an IP address.

Sample Output

```

PS C:\Users\shaya> nslookup google.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4000:101f::64
           2404:6800:4000:101f::8b
           2404:6800:4000:101f::8a
           2404:6800:4000:101f::71
           142.250.67.14

```

This command is used to find the IP address associated with the domain name google.com. It queries the Domain Name System server to translate the website's domain name into its corresponding IP address.

Server: dns.google

Address: 8.8.8.8

Name: google.com

Address: 142.250.xx.xx

Conclusion:

[These commands helped in understanding the computer's network configuration, testing internet connectivity, tracing the route taken by data packets, and resolving domain names to IP addresses. Overall, the experiment provided practical knowledge of basic network diagnostics and troubleshooting tools commonly used in computer networks.]